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A Design Scheme of STEM Digital Textbook Based on Metaverse and Generative Artificial Intelligence

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Abstract

STEM teaching involves a wealth of interdisciplinary background knowledge. For traditional digital textbooks, due to the lack of a general way to integrate STEM interdisciplinary knowledge, it is difficult to provide learners with a sense of interdisciplinary cognitive presence under the real situation, and it is difficult to be used as a cognitive scaffold for STEM interdisciplinary teaching. Based on the metaverse virtual simulation and generative artificial intelligence technology, this paper designs a STEM digital textbook system scheme, in order to realize the intelligent transformation and upgrading of STEM digital textbooks.

CCS Concepts

• Information systems;; • Information systems applications;;
• Multimedia information systems;;

Keywords

STEM education, Digital textbook, Generative artificial intelligence, Metaverse

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1 INTRODUCTION

STEM education is of great significance to cultivate interdisciplinary talents in the fields of science, technology, engineering, and mathematics. Under the trend of education digital transformation, as an important media carrying STEM education, STEM digital textbooks should not only dynamically present interdisciplinary knowledge structures of different topics in the content organization, but also provide learners with immersive embodied cognitive scaffolds in the technical representation. Therefore, how to construct STEM digital textbooks has always been a key research issue to promote the high-quality development of STEM education [1]. In the current technology wave of the vigorous development of metaverse and generative artificial intelligence, metaverse technology can realize the virtual simulation scene of STEM interdisciplinary knowledge, and generative artificial intelligence technology can realize the accurate teaching of STEM interdisciplinary teaching through the way of human-machine collaborative knowledge creation. This paper aims to combine the technical characteristics of metaverse and generative artificial intelligence to design a set of STEM digital textbooks, so that it can not only be used as an intelligent tutoring platform for human-machine collaborative creation of STEM interdisciplinary knowledge, but also as an embodied cognitive scaffold for virtual simulation and practice of STEM interdisciplinary skills. It is expected that this research can provide new solutions for the innovative design of STEM digital textbooks, and also provide useful research directions for the application exploration of metaverse and generative artificial intelligence in STEM education.

2 REVIEW OF RELATED RESEARCH WORK

2.1 The application of STEM education based on the metaverse

In 2021, the digital social giant Facebook changed its name to Meta [2], indicating that the metaverse was involved in the field of social life with a phenomenal high attitude. Subsequently, various countries issued relevant policies to support the metaverse as an important digital technology to support the transformation and development of virtual-real integration of all walks of life. For example, the European Commission stressed the importance of the

metaverse in "Building Europe's Digital Future" and China encouraged the research and development of the metaverse in "Guidelines on Accelerating the Development of Virtual Reality Industry". All indicate that the metaverse has become the key technology for the realization of virtual and real integration of digital applications in various industries around the world [3]. Based on the integration of technologies, the metaverse can realize multi-sensory interaction with virtual environments, digital objects and people [4], providing learners with an immersive digital interaction environment, which greatly enriches learners' learning experience by creating highly interactive and immersive learning scenes. In this context, embodied cognition theory plays a vital role in this framework, which emphasizes the close connection between cognitive processes and physical experience, and promotes STEM education to move beyond traditional abstract theoretical learning to a learning mode that pays more attention to practical operation and intuitive experience. In the metaverse environment, learners can interact with virtual objects through body movements, and this type of participation not only deepens the understanding of STEM concepts, but also promotes the development of innovative thinking and problem-solving skills.

The metaverse has experienced three stages of evolution in the field of education, from three-dimensional virtual world, online virtual world to the current virtual world on the chain. At present, extended reality technology, digital twin technology (DT) and blockchain technology are mainly used to provide technical support for the remodeling of education. XR-based educational metaverse has received widespread attention, and its most obvious advantage is that it can provide an immersive teaching and interaction scene for the classroom, create an environment consistent with the real world in the virtual world [5], to enhance the sense of reality, and meet the teaching and learning needs in the physical world and the virtual world [6]. The XR trend marks a new paradigm shift: the transition from physical or flat simulation to immersive, embodied scenes and environments. Among them, XR technology includes virtual reality technology (VR), augmented reality technology (AR) and mixed reality (MR) technology.

The application of XR technology in STEM education can provide real-time feedback to learners' multi-senses such as vision, hearing, and touch, so as to provide learners with an immersive and highly realistic experience. In the STEM education application, with the immersion of XR technology and other characteristics, the teaching content can have good interactive characteristics.

Through the game design based on the metaverse to teach STEM knowledge, the integration of electronic games and STEM education can achieve the effect of reducing the learning threshold. In 2018, Epic Games released a "sandbox" version of Fortnite, allowing players to modify the game's features to enhance their interest in STEM education without even knowing it. Nagaoka University of Technology (NUT) in Japan has opened a virtual STEM course based on the meta universe [7], which uses VR and AR technology to connect various real radioactive materials with radioactive materials in the virtual world. It increases the sense of reality for learners. In addition, examples such as the XR experience of offline and online courses of STEM and other disciplines in cooperation between Dreamscape Immersive and ASU, and the virtual ecological forest of Pennsylvania State University [8] also confirm that STEM

education based on the metauniverse can be achieved through adaptive learning and repeated practice. This kind of gamified learning is currently a popular STEM education method.

2.2 The STEM education application of generative artificial intelligence

At the same time, the field of generative AI is also growing rapidly, becoming increasingly important as a core concept and distinction of a rapidly evolving technology [9]. Since OpenAI launched ChatGPT in late 2022, generative AI has made significant progress in the field of text generation. ChatGPT has attracted much attention and discussion around the world due to its powerful language generation capabilities [10]. In April 2024, at the 27th United Nations Conference on Science and Technology held in Switzerland, the World Institute of Digital Technology (WDTA) released two international standards, "Security Testing Standards for Generative Artificial Intelligence Applications" and "Security Testing Methods for Large Language Models", marking that the development of generative artificial intelligence technology has been recognized and standardized by the international community. In the field of digital textbooks, with the rapid development of information technology, digital textbooks have gradually become the mainstream form in the field of education [11]. At the same time, the release of documents such as the "China Smart Education Development Report 2023" has provided a strong policy guarantee for the construction and application of digital textbooks. Generative artificial intelligence is a machine learning model based on neural network structure, which allows the machine to simulate the human learning way, extract the regularity and characteristic information from a large amount of data autonomously, and then predict and generate new content. It has the ability to generate text, speech, image and other multi-modal data. With the development of computer technology and artificial intelligence theory, the application of generative artificial intelligence in the field of education is constantly innovating and expanding, covering many aspects such as intelligent teaching materials, intelligent assessment, intelligent tutoring, and intelligent experiments.

Generative artificial intelligence can establish learner model based on learner behavior data generated in the learning process according to students' level, interest, style and other factors [12], analyze learners' thinking and ability to evaluate their learning ability, map knowledge analysis, obtain learners' knowledge mastery degree [12], and identify learners' knowledge gap. It dynamically generates or adjusts the content, difficulty, and form of STEM textbooks, and provides personalized learning resources and customized learning plans.

Generative artificial intelligence can use natural language processing, image recognition, data mining and other technologies to automatically or semi-automatically grade or give feedback to students' assignments, projects and experiments in STEM education, so as to improve the efficiency and quality of assessment [13]. At the same time, the instant scoring system based on generative artificial intelligence provides real-time feedback, which provides learners with targeted, fast, and actionable insights to help learners understand and correct errors. At the same time, it provides

teachers with detailed learning situation analysis reports to help teachers better understand learners' learning needs and progress.

AI can utilize techniques such as machine learning and knowledge representation to build models of students and domains and provide personalized guidance or interventions based on the models. The Intelligent Textbook System (ITS) for knowledge organization includes three modules: domain knowledge module, education strategy module and learner model. These modules work together to ensure that the intelligent tutoring system can provide a truly personalized and adaptive learning environment for all learners. By simulating the role of a professional tutor, ITS provides learners with customized educational content and feedback, avoiding the direct involvement of human teachers. Its core is the use of artificial intelligence to achieve highly personalized learning support, which can ensure that the system can adapt to the unique needs of each learner [14] to promote the cognitive and emotional development of students in STEM education.

Generative artificial intelligence can use VR, AR and other technologies to apply finite element virtual simulation technology to experimental teaching as a supplement to physical experiments. According to the innovative principle of "combination of virtual and real, to supplement the need", a new teaching method of "integration of theory, virtual simulation and practical operation" is realized, and realistic STEM experiment scenes and operations are provided for students. In order to enhance students' practical ability and the spirit of inquiry.

In summary, the application of generative artificial intelligence in STEM education is diverse and innovative, which provides effective technical support and solutions for the application of STEM education, and helps to realize teachers' teaching and students' learning in personalized learning.

2.3 The current research progress of digital teaching materials

With the rapid development of digital technology, digital textbooks are the inevitable trend of the development of textbooks in the digital era, and an important representation of the new prospect of education. The construction of digital textbooks has become an "accelerator" in the digital transformation of education in various countries, and is a key part of the realization of education modernization.

Japan is one of the earliest countries to develop digital textbooks. The first disk-transfer digital textbook in Japan was published in 1993. In the 21st century, the Japanese government officially started the road of education informatization and digital teaching materials construction, and successively launched two major education digitization policies: e-Japan(2001) and u-Japan (2004), and then launched i-Japan (2015). The development direction of education and talent training in the digital society is expounded. Since the beginning of construction, the construction of digital textbooks in Japan has established the dualization structure of "digital textbooks" and "digital teaching auxiliary materials" [15]. The former refers to the digital textbooks generated by digitizing the original contents of the paper textbooks. The latter refers to the video, sound, animation and other teaching materials that the textbook itself does not have. Japanese digital textbooks have an evaluation system based

on the reform of informatization teaching form, which examines students' knowledge and skills, emotional attitude, and strengthens students' learning enthusiasm and self-evaluation objectivity. The circulation system based on "personalized learning data" is one of the characteristics of the construction of digital textbooks in Japan. The system can customize a personalized and dynamically updated "knowledge map" for students according to students' learning preferences, learning performance and knowledge ability level, reflecting students' learning growth trajectory. Students can objectively evaluate themselves through the "knowledge map", review the old and learn new things, check and fill in gaps. Teachers can also understand students' learning situation and carry out personalized teaching according to the "knowledge map". The real-time update of "personalized learning data" provides fresh and flowing blood for digital textbooks, helping to realize students' personalized and efficient learning.

In 2012, Brazil explicitly required the establishment of "digital textbooks" for the first time in the "National Digital Library Plan", making it a new form of teaching material supply. In 2017, the Brazilian government launched the "Digital Education Plan", which proposed the development goal of providing digital education resources, and provided infrastructure and platform support for the development and application of digital teaching materials. In recent years, the Brazilian government has vigorously promoted the digital transformation strategy, and promoted the construction of digital textbooks through a series of network infrastructure construction projects to promote education equity and improve the quality. The concept of the construction of digital textbooks for basic education in Brazil is based on the dual drive of media information literacy and the cultivation of digital citizenship. The former emphasizes the cultivation of students' ability to obtain, analyze, evaluate, create and disseminate information in the digital space, which directly leads to the innovation and optimization of digital textbooks. The latter requires that digital culture should be regarded as a learning theme throughout all stages of basic education from kindergarten to middle school, and students should maintain independent thinking and moral judgment when using digital technology and media. In addition, the technical standards for the construction of digital textbooks in Brazil are diversified and standardized. Embedding teaching resources such as audio and video, animation and virtual simulation into digital textbooks enriches teaching forms and increases the richness and availability of teaching materials. It is worth mentioning that the Brazilian government has cooperated with the organization Discovery Education [16] to provide innovative STEM resources to improve the quality of basic education teaching.

In summary, digital textbooks from an international perspective have always integrated emerging information technology into their design and development process, in order to promote the continuous evolution of the presentation of textbook content towards a more immersive learning and interactive environment, and to make the organization of textbook knowledge more precise and intelligent. As a digital teaching resource aggregation platform that carries interdisciplinary fields such as science, technology, engineering, and mathematics, STEM digital textbooks should optimize the interdisciplinary learning experience of STEM through technological means such as metaverse and generative artificial

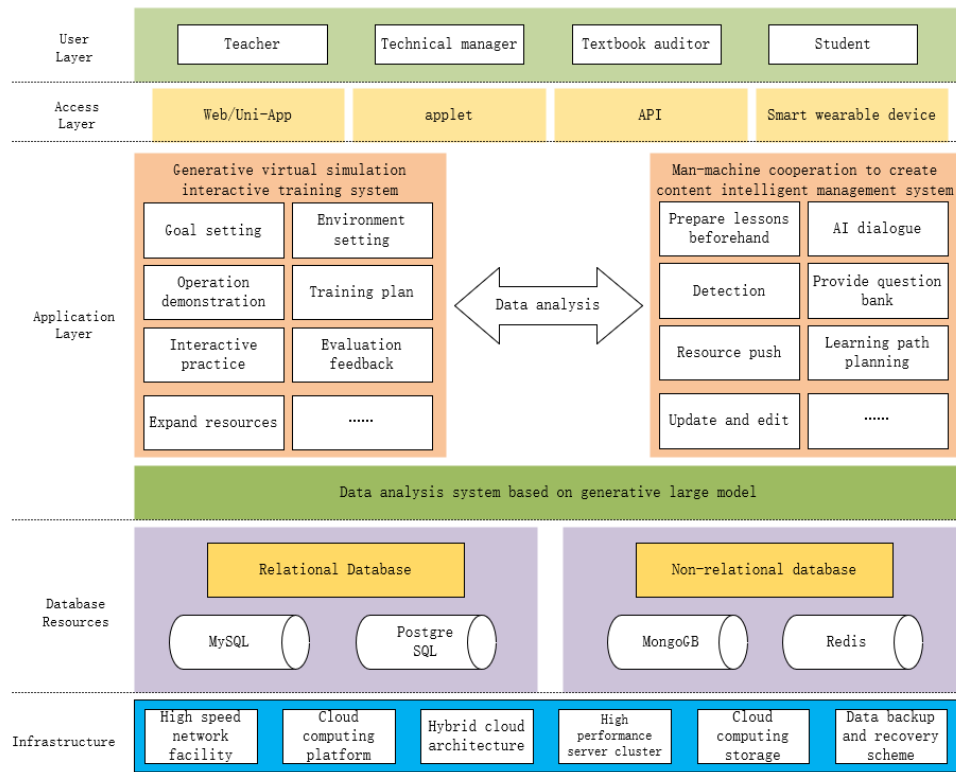


Figure 1: Schematic diagram of the overall system design

intelligence, and achieve intelligent organization of STEM interdisciplinary knowledge and resources.

3 OVERALL DESIGN OF STEM DIGITAL TEXTBOOKS

The overall design of STEM digital textbook architecture should reflect the principle of human-computer collaborative creation in the organization of teaching content, and the dynamic organization of STEM interdisciplinary knowledge resources should be driven by generative artificial intelligence to achieve accurate learning guidance. In the aspect of teaching practice experience, the principle of immersive interaction of virtual and real scenes should be reflected, and the practical virtual simulation experience of STEM interdisciplinary skills should be constructed by metaverse technology, so as to promote learners to achieve the higher-order learning goals of embodied perception and human-computer co-creation. After comprehensively considering the characteristics of metaverse, generative artificial intelligence, and the interdisciplinary nature of STEM digital textbooks, this paper designed the digital textbook system, as shown in Figure 1, which is mainly composed of three key components: content management system, data analysis system, and interactive learning system.

3.1 Human-machine collaborative content intelligent management system

Content management system plays an important role in the whole system, the main task is to manage the teaching content and all kinds of resources. Through the teaching content management system, it realizes the efficient organization and convenient management of the interdisciplinary teaching material content, and edits it at any time. The resource sharing system included in it can collect multi-party resources, promote the circulation and sharing of various resources such as situation creation and experimental simulation in different users and scenarios, form an item bank and accurately push different users, improve resource utilization and expand value. The multimodal large model technology can provide rich and diverse STEM interdisciplinary subject learning resources and realize semantic integration with the knowledge system of traditional single-subject digital textbooks, provide more appropriate interdisciplinary problem solving paths and learning evaluation mechanisms for learning, push resources and generate learning path plans for users. The resource update system can ensure that the latest STEM knowledge and research results are integrated into the textbooks in a timely manner. Jointly realize the orderly management, allocation and update of learning resources, and create a high-quality and constantly evolving learning environment for learners.

3.2 Data analysis system based on generative large model

As the core data engine of STEM digital textbooks, the data analysis system is the intelligent data training base for various STEM interdisciplinary learning topics. Its main task is to gather the teaching status data of STEM digital textbooks system in real time for teaching status analysis, and provide a basis for decision-making through human-computer collaborative interaction. To improve the intelligent level, fault tolerance and accuracy of STEM digital textbook system in teaching application. Through the learning data analysis system, it can comprehensively collect various learning behavior data of students in the platform, such as access frequency, stay time, interaction times, etc., to provide the basis for subsequent analysis. The learning effect evaluation system can deeply analyze students' learning outcomes, so as to accurately evaluate the learning effect. The personalized learning system uses the generative large model intelligent algorithm to analyze the learning progress and ability status of students in real time based on the status data such as learning annotations and learning exercise feedback in STEM digital textbooks, and intelligently recommends appropriate STEM interdisciplinary learning resources according to the cognitive principles of the zone of Recent development. It constructed a STEM interdisciplinary knowledge cognition and problem solving path that adapted to different levels of students, and gave each student accurate learning guidance and timely feedback. Its operation mode is to obtain a large amount of data through the learning data analysis system, then evaluate it by the learning effect evaluation system, and finally personalize the content recommendation by the generative big model according to the analysis results. All parts work together to provide more targeted and adaptive learning support for learners.

3.3 Generative virtual simulation interactive training system

As the learning evaluation feedback system of STEM digital textbooks, the generative virtual simulation interactive training system provides learners with intelligent training feedback and evaluation experience in the way of human-computer collaborative interaction through the generative large model embedded in the textbooks. It covers functional modules such as STEM case practice area based on generative large model and virtual reality scene training area based on metaverse, which can set goals for students, provide training plans and provide evaluation feedback. Through the virtual simulation interaction system of STEM digital textbook system, students can closely interact with each other and students and textbooks. Students can carry out in-depth interaction and real-time simulation training with learners around the world through virtual digital avatarates through various unique ways such as voting, discussion, selection, judgment, word cloud and so on. With the help of advanced metaverse technology, it constructed various real scenarios and backgrounds through the virtual interactive system, and created a highly immersive STEM learning environment, so that students could participate in activities such as experiments, design and construction in the virtual 3D scene, and feel the wonderful process of scientific exploration in person. The gamified learning system introduces an interesting gamified mechanism on

the platform. Students can successfully complete the challenge in the book situation through the form of role play, and complete various tasks and training well, which greatly improves the enthusiasm and enthusiasm of learning. The experiment simulation system strongly supports the careful design and smooth operation of the virtual experiment, and provides students with a rare experience of practical operation. The team collaboration system gives students the opportunity to create personalized digital identities, which facilitates them to form teams freely to meet difficult challenges, and exchange puzzles and experiences in the learning process. It can more effectively promote the interactive learning of students in the interdisciplinary learning process, and comprehensively improve their comprehensive ability.

4 RESOURCE INTEGRATION METHOD OF STEM DIGITAL TEXTBOOKS

Based on Generative Artificial Intelligence STEM education emphasizes interdisciplinary integration, practical operation, problem solving and the cultivation of innovative thinking ability, in order to cope with the rapidly changing scientific and technological environment and social development needs. The construction of knowledge graph plays a vital role in promoting the integration and application of knowledge between disciplines. As a structured semantic knowledge base, knowledge graph constructs an intuitive and clear knowledge concept association framework by visualizing the concepts of different disciplines and their relationships. To a certain extent, it makes up for the lack of single subject knowledge in traditional textbooks, and helps learners to deeply understand the internal connections between different subject fields. The construction of knowledge graph involves the deep mining and abstraction of subject knowledge, which not only clearly depicts the direct connections between knowledge points, but also reveals the implicit and interdisciplinary associations. The dynamic and expansibility of knowledge graph can support the update and development of subject knowledge, and provide great flexibility for the knowledge integration of digital textbooks by constantly revising and expanding to adapt to new learning needs.

The visualization of the knowledge system supported by the knowledge graph is manifested in the identification of entities in the teaching content, such as courses, knowledge points, concepts, etc. These entities constitute the basic nodes of the knowledge graph. Each entity has its specific attributes, such as the name of the course, the description of the knowledge point, the definition of the relevant concept, etc. Natural Language Processing (NLP) technology is used to automatically extract relationships from teaching materials and represent them in the form of edges in the knowledge graph, showing direct connections between entities and revealing logical and hierarchical relationships between them.

The nature of knowledge graph is heterogeneous graph. The heterogeneous structure not only reveals the connectivity and complementarity of nodes within the knowledge graph, but also uses the graph neural network to integrate the graph neural network into the information transfer and aggregation of entities in the knowledge graph, which helps to capture the potential interests of users [16]. Commonly used to extract specific entity relation knowledge in the field of education j techniques by reducing the

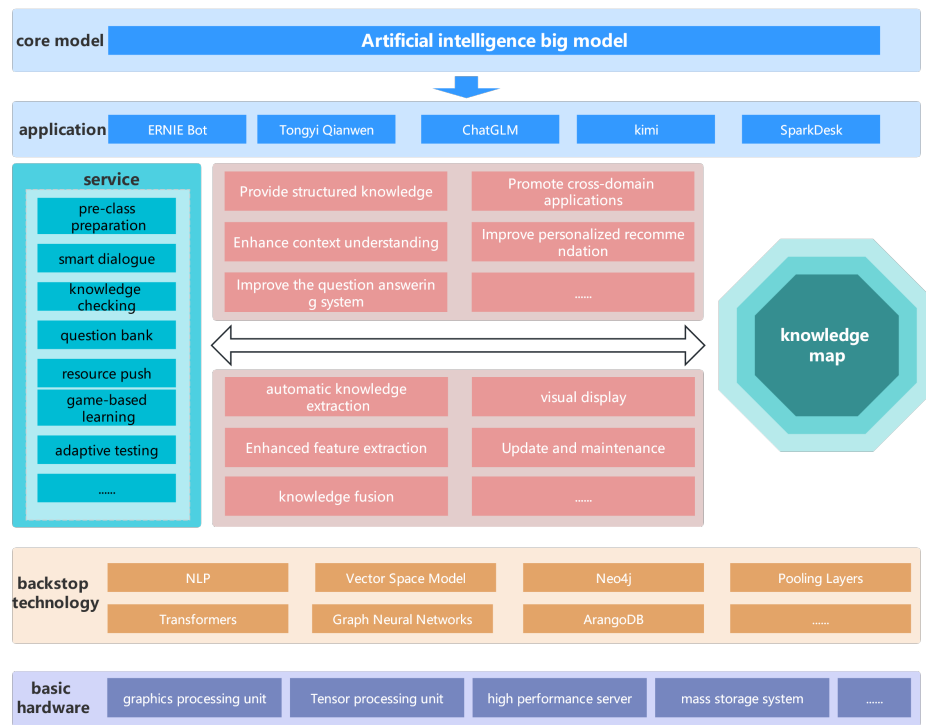


Figure 2: The method of integrating interdisciplinary knowledge resources in STEM by the generative large model

dependence on annotated resources. The Piecewise Convolutional Neural Network (PCNN) was used as the sentence encoder to assign dynamic weights to sentences with the same entity pair, and those sentences that could better convey relationships were prioritized [13], which improved the applicability of resource integration and generation effects, and constructed a high-quality knowledge graph for interdisciplinary STEM resource learning.

The intelligent textbook integrating STEM interdisciplinary knowledge resources based on the large model of generative artificial intelligence provides teachers and students with pre-class preview, intelligent question answering, knowledge detection and other services (Figure 2). The knowledge graph is generated through automatic knowledge extraction, enhanced feature extraction, knowledge fusion, visual display and other technologies, and the knowledge graph is reactive to the generative artificial intelligence. The service of intelligent textbooks is optimized by providing structured knowledge, enhancing language understanding, and improving personalized recommendation. The system supports the generation, update and maintenance of knowledge graph and intelligent textbook services through image processing unit, high-performance server, mass storage system and other auxiliary natural language processing, vector space model, Neo4j and other services.

5 VIRTUAL SIMULATION SKILLS EVALUATION METHOD OF STEM DIGITAL TEXTBOOKS BASED ON METAVERSE

As the product of the ultimate form of the future society [17], the realization of virtual simulation of digital textbooks has become the focus of discussion. As shown in Figure 3, it utilizes Virtual Reality (VR) and Augmented reality (AR) technology, 3D modeling technology, a variety of sensors and communication devices to build an unprecedented immersive learning experience for students.

VR and AR technologies bring immersive feeling to learners and unique learning opportunities for students [18]. Through head-mounted displays, gloves and other devices, VR technology constructs a 3D environment completely simulated by computers, which makes students feel as if they are in a real laboratory, historical scene or geographical environment. AR technology skillfully superimposes virtual information into the real world, so that students can interact with virtual elements in the daily environment. This combination of virtual and real learning greatly enriches the learning experience.

3D modeling technology provides realistic visual effects, educators can quickly create virtual scenes, characters and objects with three-dimensional sense, these elements not only look realistic, but also can be flexibly adjusted according to teaching needs. In such an environment, students can understand and master knowledge more intuitively.

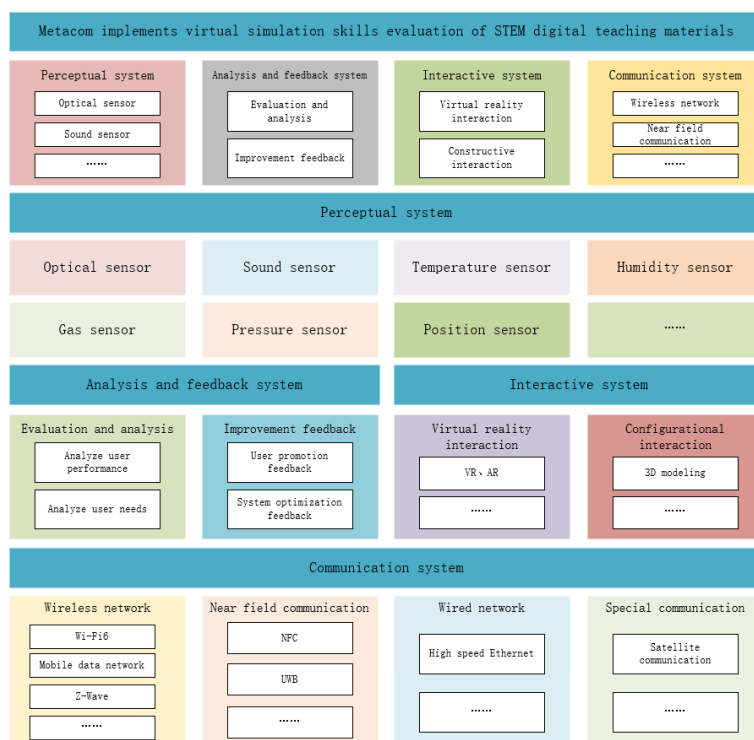


Figure 3: Virtual simulation evaluation method of interdisciplinary skills for STEM digital textbooks in metaverse

Sensors can accurately collect a variety of data, such as the user's position, motion and virtual environment related information, so as to provide a detailed basis and support for the evaluation, so that the system can accurately perceive the user's state and behavior and make corresponding responses, and can also monitor various parameters of the virtual environment in real time to maintain its authenticity and accuracy. The communication equipment is mainly responsible for the efficient and fast transmission of these data collected by the sensor, to ensure the real-time and smooth interaction between the user and the system and the system parts, and to promote the close cooperation of the system parts, so as to ensure the smooth progress of the whole virtual simulation skills evaluation process.

Metaverse provides an immersive learning experience, enabling students to understand and master knowledge more intuitively. This environment not only improves the interest and interactivity of learning, but also helps to improve the comprehensive ability of students [18]. At the same time, the recommendation of personalized learning paths meets the learning needs of different students and improves the learning efficiency. The interactive learning method encourages students to interact with the characters and objects in the virtual environment, which enhances the interest and depth of learning.

Metaverse provides strong technical support for interactive learning, enabling students to deepen their understanding in interaction with other students and with the material. At the same time, the highly immersive learning experience stimulates students' learning enthusiasm, improves students' learning motivation, reduces

teaching costs, and allows students to practice operations in the virtual environment. It provides students with a more rich, vivid and intuitive learning experience, which helps to improve their comprehensive ability.

6 IMPLEMENTATION OF A STEM DIGITAL TEXTBOOK PROTOTYPE

The STEM digital textbook platform uses the Unity 3D engine to realize the meta universe scene, and uses ChatGPT4.0, ERNIE Bot, Tongyi Qianwen and other general big model APIs as the big model interface of STEM interdisciplinary knowledge resources to realize the development of the prototype of the STEM digital textbook platform. The form of digital textbooks created through virtual simulation and human-computer co creation has increased the realism and experience of digital textbooks, and improved students' learning enthusiasm. STEM digital textbooks not only include traditional textual information, but also integrate a large number of images, videos, audio, and interactive elements, making digital educational resources more authoritative and easy to use. The main functions of the STEM digital textbook platform prototype include the integration of STEM digital textbook resources, virtual simulation of STEM teaching scenarios in the metaverse, and organization of STEM teaching resources driven by large models. The platform emphasizes student-centered approach, relying on large models to timely update and adjust teaching content, ensuring that the knowledge system of textbooks remains dynamically updated and meets the personalized learning needs of STEM education.



Figure 4: Example of Metaverse Teaching Scenarios for STEM Digital Textbooks

6.1 STEM metaverse virtual simulation teaching scene

Flipped classroom, project-based learning and other teaching models have received full support on the STEM digital textbook metaverse platform. Teachers can organize diverse virtual simulation teaching activities through the metaverse platform, allowing students to gain tangible experiences in digital practice. This textbook application mode not only improves students' participation and learning effectiveness, but also makes teaching activities more interactive and participatory. In addition, the STEM digital textbook platform has expanded the temporal and spatial scope of teaching through the metaverse virtual simulation training module. Through teaching methods such as distance education and online courses, the digital textbook metaverse platform breaks the limitations of geography and time, allowing more students to have the opportunity to connect with suitable digital educational resources. Especially for students in remote areas, they can access the same educational resources and opportunities as urban students through the digital textbook metaverse platform, further promoting educational equity. The STEM digital textbook platform, with its powerful 3D scene rendering engine, enables teachers to dynamically organize and present multimodal teaching resources such as graphics, videos, animations, etc. uploaded in real-time in the metaverse learning space, allowing students to interact with the resources on the digital textbook platform in real-time in an augmented reality manner, promoting learners' cognitive engagement and emotional participation. The metaverse virtual simulation teaching scenario in digital textbooks not only enhances students' learning interaction and enthusiasm, but also helps them consolidate knowledge and skills in virtual simulation practice, and strengthens learning effectiveness.

6.2 Organizing STEM teaching resources driven by large models

In the STEM digital textbook platform, rich multimodal teaching resources are dynamically organized through subject knowledge models. The organization of these resources is not only based on the textbook text model, but also incorporates numerous multimedia data training schemes with subject characteristics. Digital textbooks match learning themes through large models, generating interactive micro lessons, personalized question banks, and knowledge or skill point mini games. The "Yuanxi" digital textbook can also optimize the teaching scenarios of the metaverse through the resource organization ability of large models, presenting teaching activities in a gamified manner, allowing students to consolidate knowledge and skills in a relaxed and enjoyable atmosphere. At the same time, the digital textbook subject knowledge model will provide feedback and suggestions based on students' answer situations, helping them correct errors in a timely manner. The digital textbook model generates personalized intelligent teaching assistants by training students' digital notes, bookmarks, and annotations. It integrates the image of a virtual teacher into the learning scene of digital textbooks, adding a sense of teaching presence for autonomous learning using digital textbooks. The intelligent teaching assistant generated by the large model can quickly and accurately understand students' problems, locate key information through semantic analysis and intent recognition technology, and provide real-time feedback, which helps students maintain learning flow and promote the occurrence of deep learning.

7 SUMMARY

This study explores the application potential of metaverse and generative AI in the field of STEM digital textbooks and reveals their revolutionary educational opportunities. Metaverse technologies, such as virtual reality (VR), augmented reality (AR) and 3D modeling technology, have significantly improved learners' interest and engagement by building immersive and interactive learning

Thyroid hormones

Generally speaking, thyroid hormones or thyroid levels increase heart rate and contraction. The impact of thyroid hormones is usually much longer than that of catecholamine. The physiological activity form of thyroid hormones, the T3 or triiodothyronine has been proven to directly enter the cardiovascular cells and change the activity of the genetic group level. It also has a shadow that resembles the ability to react between the adrenaline of the above and the adrenaline of ne. Too high thyroid levels may cause excitement too fast.

Calcium

Calcium ion levels have a great impact on heart rate and shrinkage; as calcium ion levels increase, hr and contraction will also increase. High levels of calcium ions (high calcium blood syndrome) may be related to the short qt period and the t-wave widening in the ecg. The qt period indicates that from the beginning of the polarization to the rhythm rejuvenation time, including the contraction period of the heart chamber. The extremely high level of calcium can induce the sudden stop of the heart. Medicine known as calcium channel reservoir slows down hr by combining with these channels and blocking or slows down the inward movement of calcium ions.

Caffeine and nicotine

Caffeine and nicotine are not naturally present in the body. These two non-tangular pharmacies are generally exciting to the neural membranes, and have a stimulating action on the center of the heart, resulting in the increase of hr. Caffeine acts by increasing the erectalization rate of sa nodes, while nicotine stimulates the activity of transmitting impulsive to the heart's sexual neurons. Although caffeine is the most widely consumed spiritual active drugs in the world, it is legal and not regulated. Although the exact number has not been determined yet, "normal" consumption is considered harmless to most people, although it may cause sleep interruption and act as a urine sharpener. Pregnant women don't eat it, although there is no evidence indicating that there is a negative sound. The tolerance of the drug even addiction to the body and mind lead to individual consumption of the substance frequently.

Nicotine is also an exciting agent that will produce addiction. Although legal and unregulated, concerns about nicotine security and connection with respiratory systems and heart disease have led warning labels on cigarette packaging.

Factors with reduced heart rate

When a person has undergone changes in the level of sodium and potassium, hypoxia, acid poisoning, alkali poisoning and body temperature, the heart rate will slower (see table 19.1) The relationship between electrolyte and hr is complicated, but maintaining the balance of electrolyte is essential for normal polarization waves. In these two ions, potassium has greater clinical significance. Initially, low sodium bleeding (low sodium levels) and high sodium blood syndrome (high sodium levels) may lead to excessive heartbeat. Serious high sodium bleeding may cause tremors, resulting in carbon monoxide stopping. Serious

Is the impact of thyroid hormones too high and too low?

Both too high and too low thyroid hormones can have different effects on the body. Thyroid hormones, including thyroxine (T4) and triiodothyronine (T3), are important hormones that regulate the body's metabolism. They are secreted by the thyroid gland and regulated by thyroid stimulating hormone (TSH) secreted by the pituitary gland. When thyroid hormone is too high, known as hyperthyroidism (hyperthyroidism), it can lead to a hypermetabolic state. In this state, the body's metabolic rate is accelerated, and the following symptoms may occur :- flustered - increased sweating - shaky hands

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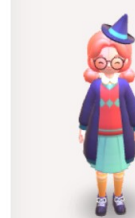


Figure 5: Example of Application Scenarios for STEM Digital Textbook Large Models

environments. Generative AI can provide personalized learning resources and plans based on students' learning data and learning styles, and can automatically evaluate learning outcomes and provide targeted feedback, thereby promoting the personalized development of learners. In addition, this paper analyzes the experiences of countries such as Japan and Brazil in the construction of digital textbooks and highlights the importance of knowledge integration and visualization in the construction of digital textbooks. Based on the above analysis, this paper proposes an innovative STEM digital textbook design scheme, which integrates three key components: content management system, data analysis system and interactive learning system, constructs a personalized, immersive and interactive learning environment, uses metaverse technology to realize the virtual simulation of digital textbooks, and constructs a realistic learning scene and experimental environment. And it uses generative artificial intelligence to realize the knowledge integration and personalized learning of digital textbooks, and provides customized learning resources and support for students. Although the field still faces challenges such as technology cost, data security, and ethical norms, with the continuous development of metaverse and generative artificial intelligence technology, STEM digital textbooks are expected to realize intelligent, personalized, and immersive development, bring more possibilities to the field of education, and create more efficient and interesting learning experiences for learners.

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